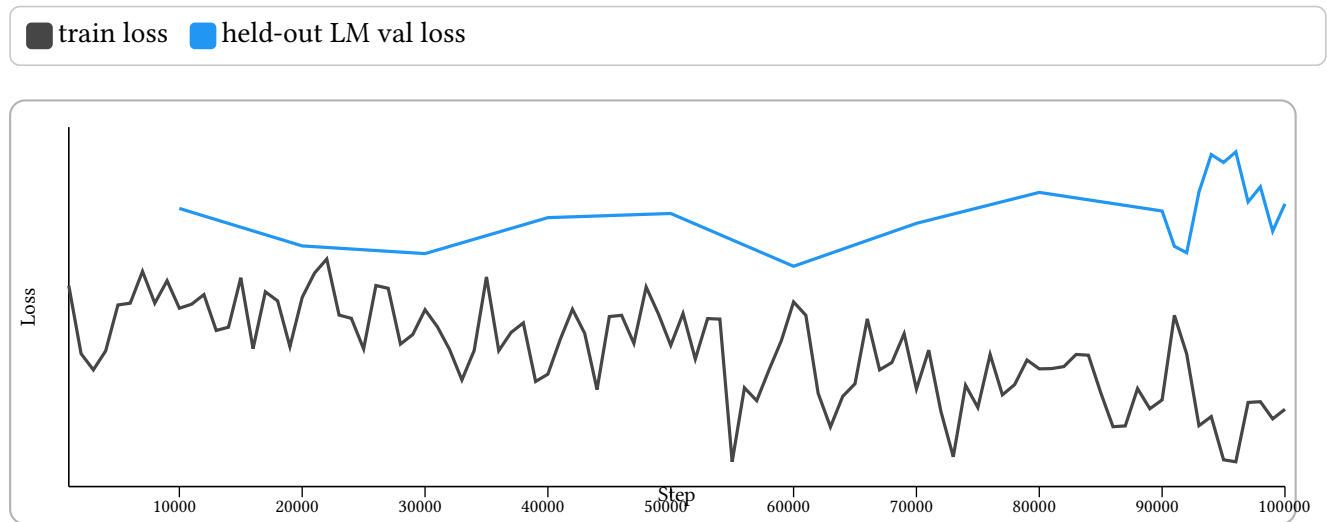


## Long Grokking Probe, full 100k

Config: long\_40m\_l1\_grokking\_probe\_100k.yml Model: 38.64M params Run: pure LM, 1x RTX 5090 Train curve from logs.txt at every 1k; held-out LM val curve from evaluated checkpoints.



- best held-out LM val loss in the sampled full run is 1.2602 at step 60000
- best train loss from the 1k samples is 0.4860 at step 55000
- val loss goes from 1.4891 at step 10000 to 1.5074 at step 100000

This still does not read like classic late grokking. Generalization improves well past the early 7k result and reaches its best sampled point around 60k, but it does not show a late monotonic jump near the end; the last 10k is worse than that 60k minimum even while train loss stays low.